

Section 1: Basic Conditional Structures

1. ****If-Else Statement****:

Write a C++ program that prompts the user to enter an integer. Your program should then determine if the entered number is positive, negative, or zero using an `if-else` statement and print the appropriate message.

2. ****If-Else If-Else Ladder****:

Create a C++ program that asks a student for their numerical exam score (an integer between 0 and 100). Based on the score, use an `if-else if-else` ladder to assign and print a letter grade according to the following scale:

- * 90-100: A
- * 80-89: B
- * 70-79: C
- * 60-69: D
- * Below 60: F

Your program should also handle invalid input (scores outside the 0-100 range) by printing an message.

Section 2: Advanced Conditional Logic and Operators

3. ****Logical Operators (`&&`, `||`, `!`)****:

Write a C++ program that takes three boolean inputs (`bool a, b, c`) from the user (e.g., as 0 or 1, then convert to `bool`). The program should then print:

- * `true` if `a` is true AND `b` is false.
- * `true` if `a` is true OR `c` is true.
- * `true` if `b` is NOT true.
- * Otherwise, print `false`.

You must use `&&`, `||`, and `!` operators appropriately.

4. ****Nested If Statements****:

A user wants to book a movie ticket. The rules are:

- * If the user is a

member: they get a 20% discount if the movie is 3D, otherwise a 10% discount.

- * If the user is a

member: they get a 5% discount only if the movie is 2D. No discount for 3D movies.

- * If the user is

: no discount.

Write a C++ code snippet that takes a `membershipType` (string) and `is3DMovie` (boolean) as input. Calculate and print the discount percentage. Use nested `if` statements.

5. ****Switch Statement****:

Develop a menu-driven program using a `switch` statement. The program should display options for unit conversion:

1. Convert Celsius to Fahrenheit
2. Convert Fahrenheit to Celsius
3. Exit

Prompt the user to choose an option. If option 1 or 2 is chosen, ask for the temperature value, perform the conversion ($F = C \times 9/5 + 32$; $C = (F - 32) \times 5/9$), and print the result. Handle invalid menu choices.

Section 3: Application and Problem Solving

6. ****Conditional (Ternary) Operator****:

Write a C++ program that prompts the user to enter their age. Use the conditional (ternary) operator to determine if the person is an

(age 18 or older) or a

(under 18). Store the result in a `std::string` variable and then print it.

7. ****Debugging Conditional Logic****:

The following C++ code snippet attempts to determine if a given year is a leap year. A leap year occurs every 4 years, except for years divisible by 100 but not by 400.

```
```cpp
#include <iostream>
#include <string>

int main() {
 int year = 1900; // Test with 1900 (not a leap year)
 // int year = 2000; // Test with 2000 (a leap year)
 // int year = 2024; // Test with 2024 (a leap year)

 std::string message;
 if (year % 4 == 0 || year % 100 != 0 && year % 400 == 0) {
 message =
;
 } else {
```

```

 message =
;
 }

 std::cout << year <<

<< message << std::endl;
 return 0;
}
```

```

Identify the logical error(s) in the `if` condition that would cause incorrect output for certain years (e.g., 1900). Explain why it's incorrect and provide the corrected C++ `if` statement.

8. **Complex Conditional Problem**:

An online store calculates shipping costs based on the total order value and customer type.

- * **Base Rule**: If the order total is \$100 or more, apply a 10% discount.

- * **Premium Customer**: If the customer is a

member, they get an *additional* 5% discount on the *already discounted price* (if any).

- * **Shipping**: After all discounts, if the final price is still over \$200, a flat \$15 shipping fee is added. Otherwise, shipping is free.

Write a C++ program that takes the `baseOrderTotal` (double) and `isPremiumCustomer` (boolean) as input. Calculate and print the `finalPrice` including any discounts and shipping fees. Show the step-by-step calculation (e.g.,

).